

```

import pandas as pd
import numpy as np

# 1. Load the fully constrained architectural space from Exercise 1
# This represents the "whittled down" full factorial matrix.
df = pd.read_csv('arch_space.csv')

# Verify the global space size is exactly 184,212
N_total = len(df)
print(f"Total valid architectures loaded: {N_total}")

target_sample_size = 90 # Target size < 100

# 2. Calculate the deterministic stride (k)
k = N_total // target_sample_size

# 3. Sort lexicographically to ensure uniform categorical grouping
# This orders the dataframe logically from D1 -> D8
columns_to_sort = list(df.columns)
df_sorted = df.sort_values(by=columns_to_sort).reset_index(drop=True)

# 4. Perform Systematic Sampling (Deterministic Grid Extraction)
deterministic_sample = df_sorted.iloc[::k].head(target_sample_size)

# 5. Format into vector representation strings
# Vector structure: D1-D2-D3-D4-D5-D6-D7-D8
vectors = []
for _, row in deterministic_sample.iterrows():
    # Concatenate D4 bits
    d4_str = f"{int(row['d4_1'])}{int(row['d4_2'])}{int(row['d4_3'])}{i
    # Concatenate D7 digits
    d7_str = f"{int(row['d7_1'])}{int(row['d7_2'])}{int(row['d7_3'])}{i

    # Build final vector
    vec = f"{int(row['d1'])} - {int(row['d2'])} - {int(row['d3'])} - {d
    vectors.append(vec)

print(f"\n--- Generated {len(vectors)} Deterministic Architectures ---"
for i, v in enumerate(vectors):
    print(f"Arch {i+1:02d}: {v}")

# Save to CSV for submission
pd.DataFrame(vectors, columns=['Vector_Representation']).to_csv('determ

```

Total valid architectures loaded: 184212

--- Generated 90 Deterministic Architectures ---

Arch 01:	0	-	0	-	0	-	000001	-	0	-	0	-	000001	-	0
Arch 02:	0	-	0	-	0	-	000100	-	1	-	1	-	012222	-	0
Arch 03:	0	-	0	-	0	-	010000	-	0	-	1	-	012221	-	0
Arch 04:	0	-	0	-	1	-	000101	-	0	-	0	-	010201	-	0
Arch 05:	0	-	0	-	1	-	001001	-	2	-	0	-	011211	-	0
Arch 06:	0	-	0	-	1	-	010001	-	1	-	0	-	011210	-	0
Arch 07:	0	-	0	-	1	-	011000	-	0	-	0	-	012220	-	0
Arch 08:	0	-	0	-	1	-	100001	-	2	-	1	-	010020	-	0
Arch 09:	0	-	0	-	1	-	101000	-	1	-	1	-	010012	-	0
Arch 10:	0	-	0	-	2	-	011101	-	0	-	1	-	010121	-	0
Arch 11:	0	-	0	-	2	-	101101	-	2	-	1	-	012212	-	0
Arch 12:	0	-	0	-	2	-	111001	-	1	-	1	-	012211	-	0
Arch 13:	0	-	1	-	0	-	000001	-	0	-	1	-	012210	-	0
Arch 14:	0	-	1	-	0	-	001000	-	0	-	0	-	010120	-	0
Arch 15:	0	-	1	-	0	-	010000	-	2	-	0	-	011200	-	0
Arch 16:	0	-	1	-	1	-	000101	-	1	-	0	-	011122	-	0
Arch 17:	0	-	1	-	1	-	001100	-	0	-	0	-	012202	-	0
Arch 18:	0	-	1	-	1	-	010001	-	2	-	1	-	001221	-	0
Arch 19:	0	-	1	-	1	-	011000	-	1	-	1	-	001220	-	0
Arch 20:	0	-	1	-	1	-	100100	-	0	-	1	-	010110	-	0
Arch 21:	0	-	1	-	1	-	101000	-	2	-	1	-	012201	-	0
Arch 22:	0	-	1	-	2	-	011101	-	1	-	1	-	012200	-	0
Arch 23:	0	-	1	-	2	-	110101	-	0	-	1	-	012122	-	0
Arch 24:	0	-	1	-	2	-	111100	-	0	-	0	-	010102	-	0
Arch 25:	0	-	2	-	0	-	000001	-	2	-	0	-	011112	-	0
Arch 26:	0	-	2	-	0	-	001000	-	1	-	0	-	011102	-	0
Arch 27:	0	-	2	-	0	-	100000	-	0	-	0	-	012121	-	0
Arch 28:	0	-	2	-	1	-	000101	-	2	-	1	-	001210	-	0
Arch 29:	0	-	2	-	1	-	001100	-	1	-	1	-	001202	-	0
Arch 30:	0	-	2	-	1	-	010100	-	0	-	1	-	010022	-	0
Arch 31:	0	-	2	-	1	-	011000	-	2	-	1	-	012120	-	0
Arch 32:	0	-	2	-	1	-	100100	-	1	-	1	-	012112	-	0
Arch 33:	0	-	2	-	1	-	110000	-	0	-	1	-	012111	-	0
Arch 34:	0	-	2	-	2	-	101101	-	0	-	0	-	010021	-	0
Arch 35:	0	-	2	-	2	-	110101	-	2	-	0	-	011020	-	0
Arch 36:	0	-	2	-	2	-	111100	-	1	-	0	-	011012	-	0
Arch 37:	1	-	0	-	0	-	000100	-	0	-	0	-	012110	-	0
Arch 38:	1	-	0	-	0	-	001000	-	2	-	1	-	001122	-	0
Arch 39:	1	-	0	-	0	-	100000	-	1	-	1	-	001121	-	0
Arch 40:	1	-	0	-	1	-	001001	-	0	-	1	-	010011	-	0
Arch 41:	1	-	0	-	1	-	001100	-	2	-	1	-	012102	-	0
Arch 42:	1	-	0	-	1	-	010100	-	1	-	1	-	012101	-	0
Arch 43:	1	-	0	-	1	-	100001	-	0	-	1	-	012100	-	0
Arch 44:	1	-	0	-	1	-	101000	-	0	-	0	-	010010	-	0
Arch 45:	1	-	0	-	1	-	110000	-	2	-	0	-	010221	-	0
Arch 46:	1	-	0	-	2	-	101101	-	1	-	0	-	010220	-	0
Arch 47:	1	-	0	-	2	-	111001	-	0	-	0	-	012022	-	0
Arch 48:	1	-	0	-	2	-	111100	-	2	-	1	-	001102	-	0
Arch 49:	1	-	1	-	0	-	000100	-	1	-	1	-	001022	-	0
Arch 50:	1	-	1	-	0	-	010000	-	0	-	1	-	010000	-	0
Arch 51:	1	-	1	-	0	-	100000	-	2	-	1	-	012021	-	0
Arch 52:	1	-	1	-	1	-	001001	-	1	-	1	-	012020	-	0

Arch 53: 1 - 1 - 1 - 010001 - 0 - 1 - 012012 - 0
Arch 54: 1 - 1 - 1 - 011000 - 0 - 0 - 001222 - 0
Arch 55: 1 - 1 - 1 - 100001 - 2 - 0 - 010210 - 0